Emanuele Chini

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Driven, innovative, and passionate about tackling challenges. Currently pursuing a National Ph.D. in Artificial Intelligence at the University La Sapienza, in collaboration with the University of Verona and Giordano Controls, with a strong passion for addressing the explainability problem in AI. Highly ambitious, cheerful, and dependable, with a strong commitment to teamwork and achieving shared goals.

Education

University La Sapienza - Rome 2023 - Present
National Doctorate Program in Artificial Intelligence *Rome, Italy*

University of Verona (110/110 cum laude) 2021 - 2023
Master's Degree in Medical Bioinformatics *Verona, Italy*

University of Trento 2018 - 2021
Bachelor's Degree in Information Engineering and Business Organization *Trento, Italy*

Experience

Explainable Knowledge Distillation in Time Series 01/2025 – 06/2025
Collaboration Research *Université Haute Alsace, Mulhouse, France*

- Research collaboration with Prof. Germain Forestier on the topic of Explainable Knowledge Distillation in Time Series, utilizing real-world datasets for practical applications.

Predictive Maintenance for HVAC Systems 11/2023 – Present
Intern *Giordano Controls, Verona, Italy*

- Developed form the ground up a pipeline that downloads data from cloud service, preprocessing them and analyse them with AI models.
- Utilized TensorFlow and PyTorch to develop custom AI algorithms and implemented state-of-the-art models from the literature.
- Achieved preliminary results in early anomaly prediction: 81% accuracy, 80% precision, and 81% recall.

Development of an Interface for OCR Data Integration with NLP Pipeline 06/2023 – 10/2023
Developer *University of Verona, Verona, Italy*

- Collaborated with Prof. Pietro Sala to develop a Flutter application connecting OCR-processed data with an NLP pipeline.

Integration of BPMN Engine in a Breast Cancer Care Management 04/2023 – 06/2023
Intern *Fondazione Bruno Kessler, Trento, Italy*

- Optimized the definition of the breast cancer care process modeled with BPMN 2.0.
- Integrated the BPMN engine between the TreC platform and the Camunda engine to enhance process management.

Publications

PACO: A Petri Net-Based Tool for Designing, Simulating, and Analyzing Multi-Objective Stochastic Processes | International Conference on Applications and Theory of Petri Nets and Concurrency (2026)

- **Accepted for publication, expected summer 2026.**
- **Abstract**: We present PACO, a tool for strategy synthesis and explanation for stochastic processes based on the BPMN 2.0 standard, extended with probabilistic splits and positive impact vectors associated with tasks. The formal semantics is provided by Synchronous Probabilistic Impactful Networks (SPIN), an enriched Petri Net model. At its core, PACO implements an on-the-fly strategy synthesis algorithm for the input SPIN. To improve transparency, PACO includes an explainer module that decomposes synthesized strategies into minimal decision trees attached to individual process choices. PACO is provided as a web-based app that supports design, synthesis, explanation, and interactive simulation of stochastic processes. To further enhance usability at design time, PACO integrates an LLM-assisted design component that helps users create and interpret processes from natural-language descriptions. Finally, tested over a synthetic dataset of processes of increasing complexity, PACO demonstrates that explainable, automated strategy synthesis is practical for realistic BPMN process models despite the inherent computational complexity of the problem.

- **Keywords**: *Strategy Synthesis*, *Multi-Objective Optimization*, *Explainable Planning*, *Markov Decision Processes*, *Probabilistic Plannin*
- **Authors**: Emanuele Chini, Daniel Amadori, Pietro Sala, Sidra Nasir Rajput, Matteo Baldi, Mattia Cappelletti

DRIFTS: Distributed Robustness for Time Series is Feasible, Distributed, and Fun | IEEE Conference on Artificial Intelligence (2026)

- **Accepted for publication, expected summer 2026.**
- **Abstract**: As complex machine learning models are increasingly integrated into critical systems, ensuring their robustness, transparency, and trustworthiness becomes essential. Explainability methods that offer formal and verifiable guarantees about model behavior are crucial not only for interpretability but also for robust decision assurance. In this work, we address the problem of generating formally grounded and robustness-aware explanations for classifiers operating in the time series domain. Our main objective is to introduce the concept of a continuous anti reason. We formalize this concept as an Interval Constraint Function (ICF) which, given a classifier and a sample, assigns an interval for each feature value of the sample. An ICF that constitutes a continuous anti reason for a given sample ensures that all data points whose feature values lie within the corresponding intervals are classified differently from the sample class. Each continuous anti reason is associated with a cost, quantifying the extent of perturbation required to induce a class change. The higher the cost of a continuous anti reason, less robust the classifier is on that sample. Therefore such a property is paramount for precisely characterizing the robustness boundaries of the classifier on a sample-wise basis. Given that computing a continuous anti reason is coNP-COMPLETE, we develop the framework called *Distributed Robustness Interval for Time Series (DRIFTS)*, a distributed algorithm that leverages an in-memory key-value store to coordinate multiple workers. The effectiveness and scalability of our approach are validated empirically on Random Forest models using univariate time series datasets from the UCR Archive.
- **Keywords**: *Sustainable and Trustworthy AI*, *Random Forest*, *Time Series*
- **Authors**: Daniel Amadori, Emanuele Chini, and Pietro Sala

Time Series Step Tree: A Novel Interpretable Method for Prompt Classification of Time Series | The 8th International Conference on the Dynamics of Information (2025)

- **Abstract**: Presented Time Series Step Tree (TSST), a novel and interpretable method for efficient time series classification using a step-wise evaluation with dynamic observation windows. TSST constructs decision trees with a unique *witness time series* selection process, ensuring both high accuracy and timely classification while maintaining transparency.
- **DOI:** 10.1007/978-3-032-16469-8_14
- **Keywords**: *Prompt time series classification*, *Interpretability*, *Decision tree*, *Univariate time series*, *Early time series classification*
- **Authors**: Omid Zare, Pietro Sala, Daniel Amadori, Emanuele Chini, and Javad Hassannataj Joloudari

Reactive Synthesis for Expected Impacts | 15th International Symposium on Games, Automata, Logics, and Formal Verification (2024)

- **Abstract**: Introduced a formal extension of BPMN incorporating choices, probabilities, and impacts (CPI) to optimize business processes. It addresses strategy synthesis to keep expected impacts within thresholds, proving PSPACE complexity, proposing an algorithm featuring automatas.
- **Code:** GitHub Repository: https://github.com/ansimonetti/PACO
- **DOI:** 10.4204/EPTCS.409.7
- **Keywords**: *Business Process Management*, *Expected Impacts*, *Reactive Synthesis*
- **Authors**: Emanuele Chini, Pietro Sala, Andrea Simonetti and Omid Zare

A Migration Framework for Active BPMN Processes in Healthcare | 12th IEEE International Conference on Healthcare Informatics (2024)

- **Abstract**: Proposed a framework for migrating BPMN processes in healthcare, integrating compensatory strategies based on users' completion status and a color-coded risk classification system. The framework is applied to an ERAS-inspired prehabilitation program for pancreatic surgery at the Verona Pancreas Institute.
- **DOI:** 10.1109/ICHI61247.2024.00050
- **Keywords**: *BPMN*, *BPMN Framework*, *Agile Methodology*, *Migration*, *Versioning*
- **Authors**: Matteo Mantovani, Emanuele Chini and Carlo Combi

Projects

BPM Based Web Application to Support Activities for Patients Waiting For Pancreatic Surgery 10/2022 - 09/2023

- Led the full-stack development of a patient support application for pancreatic surgery, utilizing Angular for the front-end and the Camunda BPM engine for the backend.
- Optimized 5 BPMN processes to enhance the application efficiency.
- Leveraged Docker for improved deployment and scalability, and optimized Client-Server synchronization to significantly enhance performance and user experience. Completed as part of my Master's thesis project.
- **Technologies:** Angular, Camunda Engine 2.0, Docker, API

Development of a Mobile Application for Verbal Apraxia Rehabilitation 06/2021-12/2021

- Collaborated with Neurocognitive Rehabilitation Center (CERIN) to develop an application for verbal apraxia rehabilitation. Acted as a liaison between the University and CERIN, facilitating effective communication and collaboration. Completed as part of my Bachelor's thesis.
- Developed an Android application from the ground up that integrates machine learning (ML) algorithms to evaluate and score mouth movements during rehabilitation exercises.
- **Technologies**: Android Studio, Google ML, Relational Database

Other Activities

Information and Computation Journal 02/2025 – Present
Reviewer

- Reviewed and evaluated 2+ articles for clarity, accuracy, and adherence to publication standards and guidelines.

Bocconi for innovation - Startup Pre-Accelerator 02/2026 – Present
Partecipant - Pyperfree

Trentino Startup Valley - Startup Accelerator 02/2026 – Present
Partecipant - Payperfree

Trentino Startup Valley - Startup Accelerator 02/2026 – Present
Partecipant - Mining Knowledge

Founders Academy 08/2024 – 10/2024
Partecipant

Technical & Soft Skills

Languages: Python, Java, TypeScript, R, Dart
Technologies: Angular, Flutter, TensorFlow, PyTorch, Flask, Docker, Android SDK, GitHub
Concepts: Artificial Intelligence, Machine Learning, Neural Networks, API, Database, Agile Methodology
Soft Skills: Teamwork, Time Management, Autonomy, Flexibility, Adaptability, Attention to Detail, Problem-Solving, Organizational Skills, and Initiative

Languages

Italian: Native – **English**: Intermediate (B2) – **German**: Basic (A2) – **French**: Basic (A2)